

Question	Answer	Marks
9(a)	using Fleming's left-hand rule force on wire is upwards	B1
	by Newton's third law, force on magnet is downwards	B1
9(b)(i)	$F = BIL$	C1
	$= 3.7 \times 10^{-3} \times 5.1 \times 8.5 \times 10^{-2}$	A1
	$= 1.6 \times 10^{-3} \text{ N}$	
9(b)(ii)	$F = 1.6 \times 10^{-3} \text{ N}$	A1
9(c)	sketch: sinusoidal wave with two cycles	B1
	amplitude $2.3 \times 10^{-3} \text{ N}$	B1
	period 0.05 s	B1

Question	Answer	Marks
10(a)	induced e.m.f. proportional to rate	M1